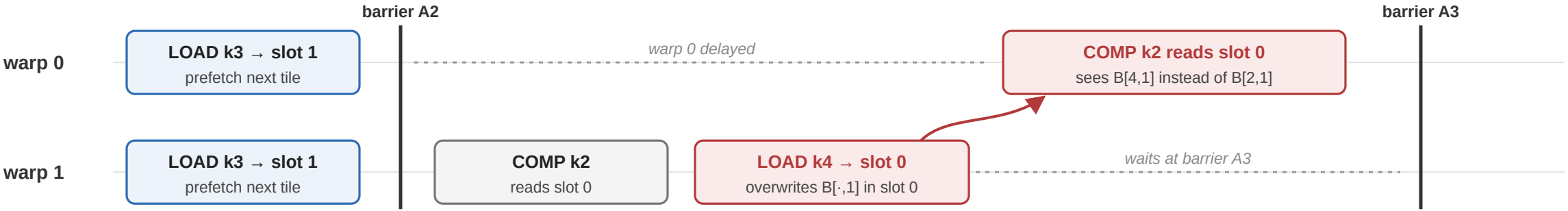
(a) S = 2, P = 1, β = 0 — admitted by the unsound precondition P ≤ S − 1

MILP predicts the same speed with less shared memory; the verification gate rejects it with this replayed interleaving



✗ C[0,1] accumulates A[0,2]·B[4,1] instead of A[0,2]·B[2,1]: one term missing, one foreign term added (lifetime obligation fails)

(b) S = 2, P = 1, β = 1 — admitted by the exact precondition P ≤ S − 2 + β

the trailing barrier B2 holds warp 1's next load until every warp has finished reading slot 0

